Rank-Optimal Directional Taylor Jets for High-Order Mixed Derivatives of Neural Surrogates

Abstract

High-order partial differential equation losses often require repeated evaluation of one prescribed mixed derivative of a neural surrogate. We develop a deterministic, algebraically exact procedure for this task. A mixed derivative is first identified with a coefficient functional of the homogeneous polynomial formed by directional Taylor coefficients. Constructing a shortest directional formula is therefore equivalent to decomposing the associated monomial into powers of linear forms. The Carlini–Catalisano–Geramita rank formula gives the minimum number of complex directions, and a roots-of-unity construction attains that minimum without solving a linear system. We evaluate each direction by Taylor-mode propagation through a multilayer perceptron and use a custom vector–Jacobian product so that derivatives of the loss with respect to network parameters remain available. Double-precision tests agree with nested automatic differentiation for both values and parameter gradients. As an application, we train complex-parameter sinh physics-informed neural networks on 12 polyharmonic, radial-chirp, and time-harmonic Maxwell settings. Under a common 1200 s, five-seed, width-128 protocol, this model has the lowest mean relative L^2 error in all 12 settings, although the margin is negligible for the hardest three-dimensional sixth-order case. The comparison is deliberately protocol-specific because trainable real parameter counts differ. The rank reduction applies to a fixed repeated-index derivative; it gives no global optimality result for general operator sums and no reduction for square-free mixed derivatives.

Keywords. Physics-informed neural network; high-order derivative; Taylor-mode automatic differentiation; Waring decomposition; apolarity; complex-valued neural network.

1 Introduction

1.1 Motivation and related work

Scientific machine-learning models frequently require derivatives of a neural surrogate with respect to its inputs. In a physics-informed neural network (PINN) [1], for example, each optimization step evaluates the differential operator in the governing equation at many collocation points. Fourth- and sixth-order equations can therefore make input differentiation a substantial part of the training cost.

A direct implementation applies reverse-mode automatic differentiation successively, retaining the graph after every coordinate derivative. The depth and storage of this nested graph grow with both the derivative order and the network depth. Three complementary strategies reduce this cost:

(i) **Forward- and Taylor-mode automatic differentiation** [2, 3], which propagates truncated Taylor coefficients through each primitive without nesting input-gradient graphs;

(ii) **Stochastic estimators** [4, 5], which write the target operator as an expectation over random tangent vectors and approximate it by sampling; and

(iii) **Operator-specific specializations** [6, 7] for the Laplacian (the trace of the Hessian), exploiting the fact that the trace collapses cleanly through linear layers.

Taylor mode supplies a derivative along any chosen direction, but does not specify a shortest set of directions for extracting one mixed partial. Stochastic methods trade work for variance, whereas trace-specialized methods address a different operator structure. Here we consider the deterministic fixed-multi-index problem and identify its directional schedule with the Waring decomposition of a monomial.

1.2 Contributions and scope

1. We prove that a directional formula for a fixed mixed derivative exists with R probes if and only if the associated degree-p monomial has a Waring decomposition with R terms. Thus the minimum probe count is a Waring rank (Section 3).

2. The Carlini–Catalisano–Geramita theorem [8] and a roots-of-unity filter give a closed-form schedule that attains the complex rank. Repeated-index patterns such as $(4, 2)$ and $(2, 2, 2)$ require, respectively, 5 rather than 7 and 9 rather than 13 directions compared with coalesced real polarization.

3. We combine this schedule with Taylor-mode propagation through smooth multilayer perceptrons and a custom vector–Jacobian product. The resulting implementation is algebraically exact, differentiable with respect to model parameters, and verified against nested automatic differentiation (Sections 4 and 5).

4. We apply the implementation to complex-parameter sinh PINNs and report five-seed, fixed-budget comparisons on polyharmonic, radial-chirp, and time-harmonic Maxwell problems (Section 6). The study records accuracy, throughput, and trainable real degrees of freedom rather than equating a common hidden width with parameter-count matching.

The rank statement is deliberately narrow. It concerns one monomial differential operator at a time. A sum such as Δ^m can be expanded and evaluated term by term; the schedule may shorten individual repeated mixed terms, but it does not exploit cancellations or shared structure across the sum. A trace-collapsing method may therefore be preferable when only a Laplacian is required. Square-free derivatives receive no rank reduction, and we make no claim of a universal wall-clock advantage over real polarization. Stochastic estimators remain preferable when a controlled approximation of a large tensor contraction is sufficient.

1.3 Notation

Throughout the paper, $u : \mathbb{R}^d \to \mathbb{R}$ denotes a sufficiently smooth scalar field. Let $\nu = (\nu_1, \dots, \nu_d) \in \mathbb{Z}^d_{\geq 0}$ be a multi-index of order $p = |\nu|$, with

$$\partial^\nu \ := \ \partial_1^{\nu_1} \cdots \partial_d^{\nu_d}. \tag{1}$$

The implementation represents the same operator by an expanded coordinate list in which coordinate k occurs ν_k times.

Relabel the *active coordinates* $\{k : \nu_k > 0\}$ as $i_0, \dots, i_n$, and set $a_j = \nu_{i_j}$. We order them so that $1 \leq a_0 \leq \cdots \leq a_n$ (ties may be broken by the original coordinate index). This relabeling singles out a minimum-exponent variable i_0, as required by the rank formula. Then

$$p \ = \ \sum_{j=0}^{n} a_j, \qquad \nu! \ = \ \prod_{k=1}^{d} \nu_k! \ = \ \prod_{j=0}^{n} a_j!, \qquad m_j \ = \ a_j + 1. \tag{2}$$

The *active subspace* V is the linear span of the active coordinate directions,

$$V \ := \ \mathrm{span}\{e_{i_0}, \dots, e_{i_n}\} \ \subset \ \mathbb{R}^d; \tag{3}$$

since ∂^ν is insensitive to coordinates outside the active set, all directional probes v below may be taken in V.

The directional Taylor coefficient of order p at x along $v \in V$ is

$$T_p(x; v) \;:=\; \frac{1}{p!}\,\frac{d^p}{d\tau^p}\,u(x+\tau v)\Big|_{\tau=0} \;=\; \frac{1}{p!}\,D^p u(x)[v,\ldots,v]. \tag{4}$$

For a complex direction $v \in V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$, we define $T_p(x; v)$ by the complex multilinear extension of $D^p u(x)$ in the right-hand side of (4). If u admits a holomorphic complexification $u_{\mathbb{C}}$, this definition equals the order-p coefficient of $u_{\mathbb{C}}(x+\tau v)$. This distinction is needed because an arbitrary C^p real function need not be defined at complex inputs. Finally, $\mathbb{K}[z]_p$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, denotes the space of degree-p homogeneous polynomials in $z = (z_0, \ldots, z_n)$.

2 Single-monomial partials as polynomial coefficient extraction

2.1 The generating polynomial

For $z = (z_0, \ldots, z_n) \in \mathbb{R}^{n+1}$ parametrizing the active subspace via $v = \sum_j z_j e_{i_j}$, the *generating polynomial* of u at x is

$$Q_u(z) \;:=\; T_p\Big(x;\, \sum_{j=0}^{n} z_j e_{i_j}\Big), \qquad z \in \mathbb{R}^{n+1}. \tag{5}$$

By construction Q_u is a degree-p homogeneous polynomial in z; the next lemma identifies its coefficients with the partial derivatives of u.

Lemma 2.1 (Generating polynomial expansion)**.** *For $u \in C^p$ on a neighborhood of x,*

$$Q_u(z) \;=\; \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!}\, z^\beta, \tag{6}$$

where β ranges over multi-indices on the active variables and $\partial^\beta = \partial_{i_0}^{\beta_0} \cdots \partial_{i_n}^{\beta_n}$.

Proof. By the symmetric multilinearity of $D^p u(x)$ and the multinomial theorem,

$$D^p u(x)\Big[\textstyle\sum_j z_j e_{i_j}, \ldots, \sum_j z_j e_{i_j}\Big] \;=\; \textstyle\sum_{|\beta|=p} \frac{p!}{\beta!}\, z^\beta\, \partial^\beta u(x).$$

Dividing by $p!$ yields (6). □

Corollary 2.2. *$\partial^\nu u(x) = \nu!\,[z^a]\,Q_u(z)$, where $a = (a_0, \ldots, a_n)$ and $[z^a]$ extracts the coefficient of $z^a = z_0^{a_0} \cdots z_n^{a_n}$.*

2.2 Directional schedules

A *directional schedule* of length R for ∂^ν is a finite collection $(c_r, v_r)_{r=1}^R \subset \mathbb{R} \times V$ such that

$$\partial^\nu u(x) \;=\; \sum_{r=1}^{R} c_r\, T_p(x; v_r), \qquad \forall\, u \in C_x^p. \tag{7}$$

Because the schedule is required to hold for every $u \in C_x^p$, the p-jet of u at x may be prescribed arbitrarily on the active variables: for any coefficients q_β, the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(x) = \beta!\, q_\beta$. This observation lets us test identities coefficient by coefficient in $\mathbb{R}[z]_p$.

The central design question of this paper is then: *What is the minimum length R^* of a directional schedule* (7)*, and how is one constructed?* Section 3 reduces this to a classical algebraic question.

3 Equivalence to monomial Waring decomposition

3.1 The bridge theorem

For $v \in V$, write $v = \sum_{j=0}^{n} v_j e_{i_j}$ in active coordinates. We pair this direction with the polynomial variable $z = (z_0, \dots, z_n)$ by $v \cdot z = \sum_{j=0}^{n} v_j z_j$.

Theorem 3.1 (Bridge)**.** *Let $R \in \mathbb{N}$, $c_r \in \mathbb{R}$, $v_r \in V$, $r = 1, \dots, R$. The following are equivalent.*

(1) *(Directional schedule)* $\partial^\nu u(x) = \sum_{r=1}^{R} c_r T_p(x; v_r) \quad \forall\, u \in C_x^p$.

(2) *(Polynomial identity in $\mathbb{R}[z]_p$)* $p!\, z^a = \sum_{r=1}^{R} c_r (v_r \cdot z)^p$.

Proof. Write $v_r = (v_{r,0}, \dots, v_{r,n})$ in active coordinates. By Lemma 2.1,

$$T_p(x; v_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!}\, v_r^\beta.$$

Substituting this expansion into statement (1), and using that the order-p active jet of u at x can be prescribed arbitrarily, shows that (1) is equivalent to the finite-dimensional coefficient system

$$\sum_{r=1}^{R} c_r\, v_r^\beta \;=\; \nu!\, \delta_{a\beta}, \qquad \forall\, \beta \in \mathbb{Z}_{\geq 0}^{n+1} \text{ with } |\beta| = p. \tag{$\star$}$$

On the other hand, expanding statement (2) gives $(v_r \cdot z)^p = \sum_{|\beta|=p} \binom{p}{\beta} v_r^\beta z^\beta$ and hence, after comparing coefficients of z^β,

$$\binom{p}{\beta} \sum_{r=1}^{R} c_r\, v_r^\beta \;=\; p!\, \delta_{a\beta}, \quad \forall\, |\beta| = p.$$

Using $p!/\binom{p}{\beta} = \beta!$ and $\beta!\, \delta_{a\beta} = \nu!\, \delta_{a\beta}$, this is exactly ($\star$). Thus (1) and (2) are equivalent. □

Definition 3.2 (Waring rank)**.** For a homogeneous polynomial $F \in \mathbb{R}[z]_p$, its *real Waring rank* is

$$R_{\mathbb{R}}(F) \;:=\; \min\Big\{ R :\; F = \sum_{r=1}^{R} c_r L_r(z)^p,\; c_r \in \mathbb{R},\; L_r \in \mathbb{R}[z]_1 \Big\}.$$

The *complex Waring rank* $R_{\mathbb{C}}(F)$ is defined identically with $\mathbb{R}$ replaced by $\mathbb{C}$; since $\mathbb{R} \subset \mathbb{C}$, $R_{\mathbb{C}}(F) \leq R_{\mathbb{R}}(F)$.

Corollary 3.3. *Among schedules with real coefficients and real directions, the minimum length R^* for ∂^ν equals the real Waring rank $R_{\mathbb{R}}(z^a)$ of the monomial.*

Remark 3.4 (Complex extension)**.** Using the complexified directional coefficient defined after (4), the same coefficient comparison holds for $c_r \in \mathbb{C}$, $v_r \in V_{\mathbb{C}}$, and identities in $\mathbb{C}[z]_p$. The minimum length of a complex schedule is therefore $R_{\mathbb{C}}(z^a)$.

3.2 The Waring rank of a monomial

The closed-form rank over $\mathbb{C}$ is classical. (Problem (P)/(Q) is posed over $\mathbb{R}$ by default; passing to $\mathbb{C}$ can only decrease the rank, so $R_{\mathbb{C}} \leq R_{\mathbb{R}}$.)

Theorem 3.5 (Carlini–Catalisano–Geramita [8])**.** *For $z^a = z_0^{a_0} \cdots z_n^{a_n}$ with $1 \leq a_0 \leq \cdots \leq a_n$,*

$$R_{\mathbb{C}}(z^a) \;=\; \prod_{j=1}^{n} (a_j + 1) \;=\; \frac{\prod_{j=0}^{n}(a_j + 1)}{a_0 + 1}. \tag{8}$$

The lower bound uses the complete-intersection apolar ideal $I_{z^a} = (\partial_0^{a_0+1}, \dots, \partial_n^{a_n+1})$; Appendix A.3 recalls the apolar interpretation. The upper bound is constructive.

Theorem 3.6 (Roots-of-unity directional schedule)**.** *Let $m_j = a_j + 1$ for $j = 1, \dots, n$, and for each tuple $\zeta = (\zeta_1, \dots, \zeta_n)$ with $\zeta_j^{m_j} = 1$ define*

$$v_\zeta = e_{i_0} + \sum_{j=1}^{n} \zeta_j e_{i_j}, \qquad c_\zeta = \frac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j. \tag{9}$$

Then $\partial^\nu u(x) = \sum_\zeta c_\zeta \, T_p(x; v_\zeta)$ for every $u \in C_x^p$. The schedule has length $\prod_{j=1}^n m_j = R_{\mathbb{C}}(z^a)$ and is therefore length-optimal over $\mathbb{C}$.

The proof, given in Appendix A.2, verifies condition (2) of Theorem 3.1 directly via the roots-of-unity filter $\sum_{\zeta^m=1} \zeta^k = m\, \mathbf{1}_{m|k}$.

To illustrate how Theorem 3.1 couples the analytic side (7) (problem (P)) with the algebraic side $p!\, z^a = \sum_r c_r (v_r \cdot z)^p$ (problem (Q)), we work out two examples: a low-order one for which both sides reduce to hand-checkable identities, and a higher-order repeated-index one that exhibits the full strength of the complex Waring construction.

Example 3.7 ($p = 2$, $a = (1, 1)$, real schedule of length 2)**.** For $a = (1, 1)$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, $U_2 = \{+1, -1\}$. Equation (9) produces the schedule

$$\big(c_{+1}, v_{+1}\big) = \big(\tfrac{1}{2},\, e_1 + e_2\big), \qquad \big(c_{-1}, v_{-1}\big) = \big(-\tfrac{1}{2},\, e_1 - e_2\big). \tag{10}$$

Verification on the analytic side. Expanding the directional Taylor coefficient $T_p(x; v) = \frac{1}{2}\big(u_{11}v_1^2 + 2u_{12}v_1v_2 + u_{22}v_2^2\big)$,

$$\tfrac{1}{2}T_p(x; e_1 + e_2) - \tfrac{1}{2}T_p(x; e_1 - e_2) = \tfrac{1}{4}\big[(u_{11} + 2u_{12} + u_{22}) - (u_{11} - 2u_{12} + u_{22})\big] = u_{12}(x).$$

Verification on the algebraic side. The equivalent polynomial identity in $\mathbb{R}[z]_2$ reads

$$2!\, z_0 z_1 \;=\; \tfrac{1}{2}(z_0 + z_1)^2 - \tfrac{1}{2}(z_0 - z_1)^2 \;=\; \tfrac{1}{2}\big[(z_0^2 + 2z_0z_1 + z_1^2) - (z_0^2 - 2z_0z_1 + z_1^2)\big] \;=\; 2\, z_0 z_1,$$

matching $p!\, z^a = 2\, z_0 z_1$. The two checks coincide via the bridge of Theorem 3.1.

Example 3.8 ($p = 6$, $a = (2, 2, 2)$, complex schedule of length 9)**.** Consider $u_{112233} = \partial_1^2\partial_2^2\partial_3^2 u$, i.e. $a = (2, 2, 2)$. Here $n = 2$, $a_0 = a_1 = a_2 = 2$, $m_1 = m_2 = 3$; take $\omega = e^{2\pi \mathrm{i}/3}$ and $U_3 = \{1, \omega, \omega^2\}$. Equation (9) produces $R_{\mathbb{C}}(z^{(2,2,2)}) = 9$ directions: for each $(\zeta_1, \zeta_2) \in U_3 \times U_3$,

$$v_\zeta \;=\; e_1 + \zeta_1 e_2 + \zeta_2 e_3, \qquad c_\zeta \;=\; \frac{(2!)^3}{3 \cdot 3}\, \zeta_1\zeta_2 \;=\; \frac{8}{9}\, \zeta_1\zeta_2. \tag{11}$$

Analytic side. The directional schedule reads

$$u_{112233}(x) \;=\; \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1\zeta_2\, T_p\big(x;\, e_1 + \zeta_1 e_2 + \zeta_2 e_3\big). \tag{12}$$

Algebraic side. The equivalent polynomial identity in $\mathbb{C}[z]_6$ is

$$6!\, z_0^2 z_1^2 z_2^2 \;=\; \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1\zeta_2\, (z_0 + \zeta_1 z_1 + \zeta_2 z_2)^6. \tag{13}$$

That (13) holds is verified by the roots-of-unity filter (Lemma A.1): expanding the right-hand side multinomially and collecting the coefficient of $z_0^{\beta_0} z_1^{\beta_1} z_2^{\beta_2}$ reduces to the product $\prod_{j=1}^{2} \sum_{\zeta_j \in U_3} \zeta_j^{1+\beta_j}$, which is non-zero exactly when $\beta_j \equiv 2 \pmod 3$ for $j = 1, 2$; together with $\sum_j \beta_j = 6$ this forces $\beta = (2, 2, 2)$, and the leading constant $\frac{8}{9} \cdot \frac{6!}{2!\,2!\,2!} \cdot 9 = 720 = 6!$ gives the right normalisation.

By comparison, the coalesced antipodally merged real polarization schedule used in our baseline requires 13 distinct nonzero directions for the same $u_{112233}(x)$; the complex Waring schedule above attains the complex Waring-rank lower bound $R_{\mathbb{C}} = 9$ (Theorem 3.5).

3.3 Pattern taxonomy

Table 1 lists the rank values for representative active-exponent patterns up to $p = 6$, comparing the complex Waring rank $R_{\mathbb{C}}$ of Theorem 3.5 with the post-antipodal polarization count 2^{p-1}, the latter being the direction count attainable by classical real polarization for a generic multi-index of order p.

pattern	example multi-index	$R_{\mathbb{C}}$	polarization
(p)	$\partial_1^p u$	1	$\leq 2^{p-1}$
$(2,1)$	u_{112}	3	3
$(1,1,1)$	u_{123}	4	4
$(3,1)$	u_{1112}	4	4
$(2,2)$	u_{1122}	**3**	4
$(2,1,1)$	u_{1123}	6	6
$(1,1,1,1)$	u_{1234}	8	8
$(4,2)$	u_{111122}	**5**	7
$(2,2,2)$	u_{112233}	**9**	13
(1^6)	u_{123456}	32	32

Table 1: Bold entries mark patterns where the complex Waring schedule is strictly shorter than the polarization count. The square-free pattern (1^p) saturates the lower bound 2^{p-1}; the method of this paper offers no improvement there.

4 Taylor-mode evaluation and backend dispatch

4.1 The Taylor jet for Linear–sinh MLPs

Given a single direction v, Theorem 3.6 reduces the problem to evaluating $T_p(x; v)$. We use forward-mode Taylor propagation [2, 3], replacing each intermediate activation $y \in \mathbb{K}^B$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, by a length-$(p+1)$ tuple $(y_0, \ldots, y_p)$ representing its truncated Taylor coefficients $y_k = (1/k!)\, d^k y/d\tau^k|_{\tau=0}$.

For an MLP $u = \phi_L \circ A_L \circ \cdots \circ \phi_1 \circ A_1$ with affine A_ℓ and entrywise activations ϕ_ℓ, the jet rules are as follows. The affine layer $y = Wx + b$ propagates each jet order independently:

$$y_0 \;=\; Wx_0 + b, \qquad y_k \;=\; Wx_k \quad (k \geq 1). \tag{14}$$

The tanh activation, with auxiliary jet $z = 1 - y^2$, satisfies

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; -\sum_{j=0}^{k} y_j\, y_{k-j}, \qquad k \geq 1. \tag{15}$$

The sinh activation, with auxiliary jet $z = \cosh(x)$, satisfies the coupled recurrence

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, y_j\, x_{k-j}, \qquad k \geq 1. \tag{16}$$

The sinh / cosh recurrence has the same coupled form as the sin / cos recurrence, but the cosh update has no sign flip. These recurrences are algebraic identities and can be evaluated on complex tensors. We use sinh for complex-parameter networks because sinh and cosh are entire. The implementation also contains sin and sigmoid recurrences for the benchmark architectures. Although the local jet of tanh is well defined away from a pole, tanh is meromorphic rather than entire.

Each Taylor coefficient y_k is a regular tensor, and the jet propagation does not nest input-gradient graphs. It stores $p+1$ coefficient tensors per activation. We additionally use a custom `torch.autograd.Function` for each activation jet, with the closed-form backward

$$g_x^{(m)} \;=\; \sum_{j=0}^{p-m} \overline{q_j}\, g_y^{(m+j)}, \qquad q(\tau) \;=\; \phi'(x(\tau)), \tag{17}$$

where q_j is the jth Taylor coefficient of q. This vector–Jacobian product avoids reverse differentiation through the intra-jet recurrence.

4.2 Backends

We compare three implemented deterministic backends for evaluating one fixed $\partial^\nu u(x)$. All three are differentiable in the network parameters θ, so each can be used directly as the residual term of a PINN loss. The implementation also provides an `auto` dispatch option, but it is a selection rule between B2 and B3 rather than a separate mathematical backend.

(B1) Direct nested AD. The reference baseline performs p successive calls to PyTorch's `autograd.grad`, each with retained computation graph. This induces a reverse-mode autograd graph of depth $O(pL)$ for a depth-L network.

(B2) Real polarization with Taylor jet. The classical antipodally merged polarization identity is

$$\partial^\nu u(x) \;=\; \frac{1}{2^p} \sum_{\epsilon\in\{\pm1\}^p} \Big(\textstyle\prod_{r=1}^p \epsilon_r\Big)\, T_p\Big(x;\, \textstyle\sum_{r=1}^p \epsilon_r\, e_{i_r}\Big). \tag{18}$$

After antipodal merging this yields at most 2^{p-1} real directions. The directional Taylor coefficients $T_p(x;\,\cdot\,)$ are evaluated by the jet propagation of Section 4.

(B3) Complex Waring with Taylor jet. The roots-of-unity construction of Theorem 3.6 produces $R_{\mathbb{C}}(z^a) = \prod_{j=1}^n (a_j+1)$ directions in $\mathbb{C}^{n+1}$; for several repeated-index patterns this is smaller than the coalesced polarization count of (B2). For real-parameter networks, the complex directions induce differentiable casts of the linear weights via `Tensor.to`; for complex-parameter networks no cast is required and the network itself acts on the complex directions natively.

4.3 The end-to-end algorithm

Algorithm 1 summarizes the complete pipeline of backend (B3): given a smooth model u (an MLP whose primitives are covered by the jet rules (14)–(16)), a point x, and a multi-index ν of order p, it returns the value $\partial^\nu u(x)$ without nested input differentiation. The implementation batches all schedule directions in one jet pass. Backend (B2) follows the same overall structure with the Waring schedule (lines 3–7) replaced by the polarization schedule (18).

Remark 4.1. By Theorem 3.1 together with Theorem 3.6, Algorithm 1 returns $\partial^\nu u(x)$ algebraically exactly; floating-point evaluation introduces round-off. The batch contains $|\mathcal{S}| = R_{\mathbb{C}}(z^a)$ directions, the minimum possible for a complex pure-power formula (Theorem 3.5).

4.4 Cost and numerical considerations

For a fixed direction, the Taylor jet stores $p+1$ coefficient tensors per activation and applies each linear map to the stacked coefficients; there is no chain of nested reverse-mode differentiation through the input coordinates. The total value-evaluation cost of B3 is therefore proportional to $R_{\mathbb{C}}(z^a)$ batched directional jets, while B2 scales with the coalesced polarization count R_{pol}. The rank reduction is an algebraic cost reduction; the realized wall-clock time also depends on complex arithmetic, batching, activation kernels, and whether parameter gradients are required.

Algorithm 1 Single-monomial $\partial^\nu u(x)$ using a complex Waring schedule and a batched Taylor jet.

Require: smooth scalar field u; point $x \in \mathbb{R}^d$; multi-index ν of order p
Ensure: the value $\partial^\nu u(x)$
relabel the active coordinates of ν as $i_0, \ldots, i_n$ with exponents $1 \leq a_0 \leq \cdots \leq a_n$
$m_j \leftarrow a_j + 1$ for $j = 1, \ldots, n$; set $\mathcal{I} \leftarrow U_{m_1} \times \cdots \times U_{m_n}$
initialise schedule $\mathcal{S} \leftarrow \emptyset$
for all $\zeta = (\zeta_1, \ldots, \zeta_n) \in \mathcal{I}$ **do**
  $v_\zeta \leftarrow e_{i_0} + \sum_{j=1}^{n} \zeta_j \, e_{i_j}$
  $c_\zeta \leftarrow \dfrac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j$
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{(c_\zeta, v_\zeta)\}$
end for
stack the directions into $V_{\mathcal{S}} \in \mathbb{C}^{|\mathcal{S}| \times d}$
form the batched input jet $\mathbf{J}_0 \leftarrow (x \text{ replicated}, V_{\mathcal{S}}, 0, \ldots, 0)$
$\mathbf{J}_L \leftarrow$ propagate $\mathbf{J}_0$ through u using (14)–(16)
$t \leftarrow [\mathbf{J}_L]_p$ ▷ one order-p coefficient per direction
return $\sum_{r=1}^{|\mathcal{S}|} c_r t_r$

The schedule coefficients are roots of unity multiplied by $\nu!/R_{\mathbb{C}}$, so coefficient magnitudes are explicit and do not come from solving an ill-conditioned linear system. The method has no finite-difference step size and no Monte Carlo variance. Remaining errors are ordinary floating-point round-off from Taylor recurrences, complex arithmetic, and the final summation over directions.

5 Implementation verification

We verify separately the algebraic schedule, derivative values, and parameter gradients. All checks use double precision. First, the roots-of-unity coefficient filter is evaluated against every degree-four monomial in five variables for the active-exponent pattern $(2, 1, 1)$; the maximum admissible absolute coefficient error is 10^{-12}. The direction counts for all patterns in Table 1 are also checked against the closed-form rank and the coalesced polarization construction.

Second, B2 and B3 are compared with B1 on real-parameter networks using tanh, sigmoid, sin, and sinh activations. The tests cover the patterns $(2, 2)$, $(1, 1, 1)$, and (3). Values must agree with nested automatic differentiation to relative tolerance 2×10^{-10} and absolute tolerance 2×10^{-11}; the imaginary remainder for a real target is bounded by 2×10^{-11}.

Third, gradients of a squared derivative loss with respect to every network parameter are compared between the jet backends and nested differentiation. The tolerance is 2×10^{-9} relative and 2×10^{-11} absolute for both real- and complex-parameter networks. Additional analytic checks cover orders $1, 2, 3, 4$, and 6 for a scalar complex sinh network. These tests establish implementation consistency, but they do not convert a direction-count reduction into a hardware-independent timing claim. The application study below therefore reports measured throughput together with accuracy.

6 Numerical experiments on high-order and complex-valued PDEs

This section evaluates the Taylor-jet implementation in complete PINN workflows. The objective is not to time an isolated derivative call, but to measure the accuracy reached under a fixed compute budget when the derivative backend is embedded in training. We compare a complex-parameter sinh network with SIREN [9], a multiscale Fourier-feature PINN (mFF-PINN) [10, 11], and MscaleDNN-2-sin [12]. The three baseline implementations were traced to pinned upstream commits and tested for architecture and derivative fidelity.

6.1 Shared protocol

Protocol `jsc_v2` uses a 1200-second wall-clock budget per (problem, method, seed), five seeds, four trainable hidden layers, 4096 fixed interior points, 512 fixed boundary points, paired per-seed collocation, and a common held-out set of 8192 points. Every hidden layer has literal width $H = 128$. Width is not a parameter-count constraint: each complex parameter counts as two real degrees of freedom, frozen Fourier maps are excluded, and a split-real Maxwell model contains two width-128 networks. The resulting counts are reported with each table.

All methods use Adam with initial learning rate 10^{-3}, cosine decay to 10^{-4}, and the same residual normalization and boundary penalty within an atomic task. The four methods receive the same collocation points for a given seed. Complex sinh uses complex128 parameters; real targets are fitted through its real output component, with a 10^{-6} penalty on imaginary parameters. For Maxwell, the full complex output is fitted and each real baseline represents real and imaginary components with separate networks.

Architecture-specific frequencies. For polyharmonic problems, the Complex sinh first-layer scale is 2π, while the mFF-PINN branches use scales 1 and π. For chirp and Maxwell problems, these become $\max(10, 2\pi a)$ and $(1, \max(2, \pi a))$, respectively. SIREN retains its upstream first- and hidden-layer frequencies of 30. MscaleDNN-2-sin uses three independent subnets at fixed input scales $1, 2$, and 4.

Derivative backend and metric. All four architectures use B3 for every monomial derivative. Scaled activations and branched embeddings are admitted only after their input derivatives and parameter gradients agree with direct automatic differentiation. Accuracy is the held-out relative L^2 error

$$\mathcal{E}_{L^2} = \frac{\|u_\theta - u_\star\|_{L^2(\Omega)}}{\|u_\star\|_{L^2(\Omega)}}. \tag{19}$$

Tables report the arithmetic mean and sample standard deviation over five seeds; history panels show the geometric mean and seedwise range. An atomic task enters the manuscript only after a validator confirms all 20 method–seed rows, finite metrics, monotone histories, width and protocol metadata, and a fixed evaluation rule. All runs used an NVIDIA H20 GPU with CUDA 12.1; the stored bundles record PyTorch 2.4.1 or 2.5.1, so throughput is compared within an atomic task rather than across tasks.

6.2 Polyharmonic benchmark

For $d \in \{2, 3\}$, let

$$u_\star(x) = \prod_{j=1}^{d} \sin(\pi x_j), \qquad \Delta^m u_\star = (-d\pi^2)^m u_\star \quad \text{in } (-1,1)^d. \tag{20}$$

We use orders $2m \in \{2, 4, 6\}$ and enforce the Navier targets $u = u_\star$ and $\Delta^j u = (-d\pi^2)^j u_\star$, $j = 1, \dots, m-1$, on the boundary; all these targets vanish for the chosen $u_\star$. Expanding Δ^m produces $\binom{d+m-1}{m}$ monomial terms. Each is evaluated independently; mixed terms such as $\partial_x^2 \partial_y^2 u$ use the Waring schedule, while pure coordinate powers need one complex direction.

In two dimensions, the mean error of Complex sinh increases from 3.36×10^{-4} at order two to 1.01×10^{-1} at order six, but remains below the three real baselines throughout. The gap

Table 2: Polyharmonic, d=2: mean ± sample standard deviation of the relative L^2 error over five seeds after 1200 s. The lowest mean in each row is bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.36} \pm \mathbf{1.57} \times 10^{-4}$	1.00 ± 0.00	$4.88 \pm 2.73 \times 10^{-3}$	$1.18 \pm 0.24 \times 10^{-2}$
4	$\mathbf{8.02} \pm \mathbf{1.86} \times 10^{-3}$	1.00 ± 0.00	$8.18 \pm 1.29 \times 10^{-1}$	$5.74 \pm 0.45 \times 10^{-1}$
6	$\mathbf{1.01} \pm \mathbf{0.53} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.02	$9.39 \pm 0.11 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

Table 3: Polyharmonic, d=3: mean ± sample standard deviation of the relative L^2 error over five seeds after 1200 s. The lowest mean in each row is bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.29} \pm \mathbf{0.39} \times 10^{-3}$	1.00 ± 0.00	$4.61 \pm 1.39 \times 10^{-1}$	$2.33 \pm 0.25 \times 10^{-1}$
4	$\mathbf{1.57} \pm \mathbf{0.73} \times 10^{-1}$	1.00 ± 0.00	$9.95 \pm 0.04 \times 10^{-1}$	$9.38 \pm 0.35 \times 10^{-1}$
6	$\mathbf{9.82} \pm \mathbf{0.09} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.01	1.07 ± 0.07

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,354; SIREN=50,177; mFF-PINN=66,305; MscaleDNN-2-sin=150,531.

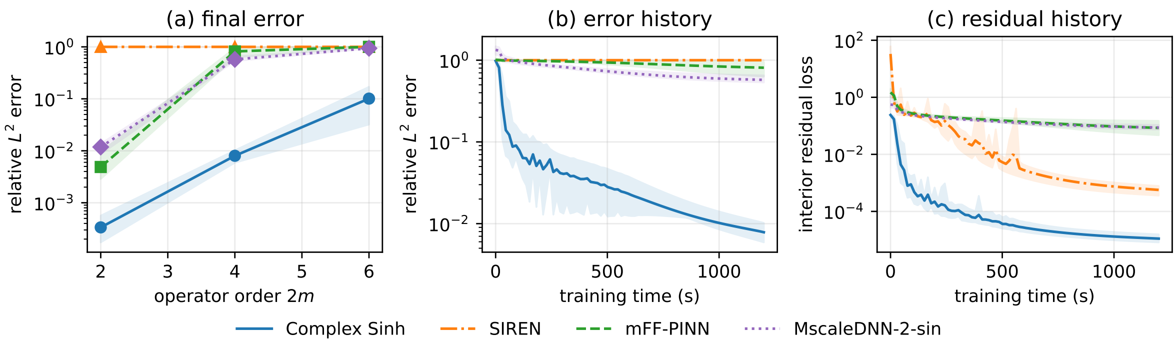

Figure 1: Two-dimensional polyharmonic benchmark. Panel (a) shows the five-seed mean relative L^2 error and seedwise range after 1200 s. Panels (b) and (c) show error and normalized interior residual histories for the representative fourth-order task.

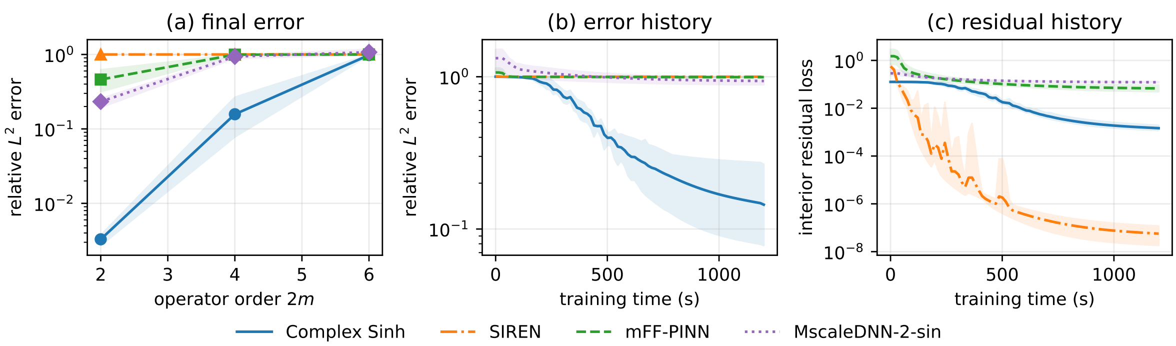

Figure 2: Three-dimensional polyharmonic benchmark, with the same convention as Fig. 1. History panels use the fourth-order task.

is largest at order four, where its mean error is approximately 72 times smaller than the next lowest mean. In three dimensions the same ordering holds at orders two and four. At order six, however, Complex sinh reaches 9.82×10^{-1}, SIREN reaches approximately 1, and the other baselines are also near unit error. This setting demonstrates budget-limited saturation rather than a practically meaningful separation.

6.3 Non-separable radial chirp

On $(-1, 1)^2$, define

$$u_\star(x, y) = \sin\left(\frac{a\pi}{2}(x^2 + y^2)\right), \qquad -\Delta u + u = f, \tag{21}$$

with Dirichlet data from $u_\star$ and $a \in \{1, 2, 3\}$. Writing $\phi = a\pi(x^2 + y^2)/2$ and $r^2 = x^2 + y^2$, the manufactured source is

$$f = (a\pi)^2 r^2 \sin\phi - 2a\pi\cos\phi + \sin\phi. \tag{22}$$

Unlike a separable Fourier mode, this target has local radial wavenumber $|\nabla\phi| = a\pi r$, so its frequency varies over the domain.

Table 4: Radial chirp: mean ± sample standard deviation of the relative L^2 error over five seeds after 1200 s. The lowest mean in each row is bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
1	$\mathbf{8.14 \pm 7.00} \times 10^{-4}$	$4.25 \pm 0.06 \times 10^{-1}$	$2.56 \pm 1.84 \times 10^{-3}$	$1.24 \pm 0.24 \times 10^{-2}$
2	$\mathbf{2.39 \pm 0.57} \times 10^{-3}$	1.53 ± 0.02	$9.69 \pm 1.00 \times 10^{-1}$	$8.47 \pm 1.85 \times 10^{-1}$
3	$\mathbf{1.00 \pm 0.18} \times 10^{-1}$	1.07 ± 0.02	1.42 ± 0.07	$9.02 \pm 0.97 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

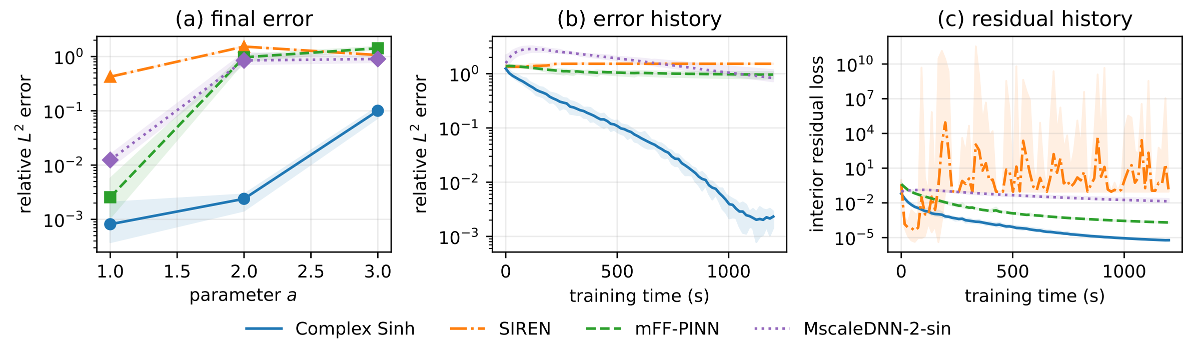

Figure 3: Radial-chirp benchmark. Panel (a) reports final five-seed accuracy and seedwise ranges. Panels (b) and (c) show accuracy and normalized residual histories for $a = 2$.

Complex sinh gives the lowest mean error at each chirp rate. At $a = 1$, its mean 8.14×10^{-4} error is about three times lower than mFF-PINN, the next method. The separation grows at $a = 2$, where the three real baselines have mean errors between 0.847 and 1.53, while Complex sinh remains at 2.39×10^{-3}. At $a = 3$, its error rises to 0.100, but remains below the real baselines. Because all methods share one loss and optimizer protocol, these results compare end-to-end methods; they do not by themselves identify representation as the sole cause of the gap.

6.4 Time-harmonic Maxwell benchmark

The two-dimensional transverse-magnetic problem is represented by

$$\Delta E + \kappa^2 E = f, \qquad \kappa^2 = (a\pi)^2(1 + \mathrm{i}\beta), \qquad E_\star = \exp\big(\mathrm{i}a\pi(x + y)\big), \tag{23}$$

with $\beta = 0.2$, exact Dirichlet data, and $a \in \{2, 4, 6\}$. The source is

$$f = \big[-2(a\pi)^2 + \kappa^2\big]E_\star.$$

Complex sinh represents E natively; each real baseline uses a split real–imaginary pair of two width-128 networks.

Table 5: Time-harmonic Maxwell: mean $\pm$ sample standard deviation of the relative L^2 error over five seeds after 1200 s. The lowest mean in each row is bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{4.21 \pm 1.57} \times 10^{-3}$	1.00 ± 0.00	$6.93 \pm 1.86 \times 10^{-1}$	1.09 ± 0.06
4	$\mathbf{4.19 \pm 0.22} \times 10^{-2}$	1.00 ± 0.00	1.06 ± 0.11	1.29 ± 0.07
6	$\mathbf{1.20 \pm 0.17} \times 10^{-1}$	1.00 ± 0.00	1.13 ± 0.15	1.34 ± 0.04

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=100,098; mFF-PINN=132,610; MscaleDNN-2-sin=300,294.

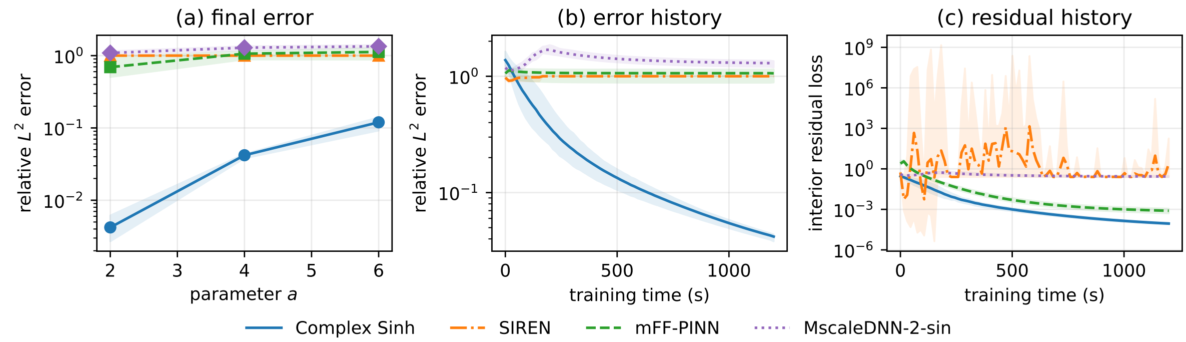

Figure 4: Time-harmonic Maxwell benchmark. Panel (a) reports final five-seed accuracy and seedwise ranges. Panels (b) and (c) show accuracy and normalized residual histories for $a = 4$.

The native-complex network has the lowest mean error at every wavenumber. Its mean error increases from 4.21×10^{-3} at $a = 2$ to 0.120 at $a = 6$. The split-real baselines remain near or above unit error at the two higher wavenumbers. This experiment favors a native complex representation structurally, but the comparison still includes architecture-dependent optimization and throughput effects; it should not be interpreted as a controlled ablation of complex arithmetic alone.

6.5 Throughput and aggregate outcome

The representative throughput rows show why equal wall-clock time is a more informative control than equal optimizer steps. On the scalar chirp task, SIREN completes the fastest steps, whereas the branched mFF-PINN and the three MscaleDNN subnets are slower. For Maxwell, each split-real baseline evaluates two networks, increasing its per-step cost. The polyharmonic row also shows the steep cost of propagating fourth-order jets through branched architectures.

Across the 12 atomic tasks, Complex sinh attains the smallest five-seed mean error in every table. In 11 settings, the next-lowest mean is at least three times larger. The exception

Table 6: Representative optimizer throughput: mean $\pm$ sample standard deviation of milliseconds per step over five seeds. Compare methods within a row; stored software versions differ between atomic tasks.

task	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
Polyharmonic, $d = 2$, $2m = 4$	243.1 ± 0.2	243.7 ± 0.2	576.8 ± 0.2	729.7 ± 0.4
Radial chirp, $a = 2$	70.7 ± 0.2	61.9 ± 0.0	145.2 ± 0.1	182.3 ± 0.0
Maxwell, $a = 4$	70.8 ± 0.1	123.5 ± 0.0	290.1 ± 0.0	367.8 ± 0.1

is the three-dimensional sixth-order polyharmonic task, where all methods are close to the trivial-predictor error. This distinction is retained in the discussion rather than reducing the benchmark to an unqualified win count.

7 Discussion and limitations

7.1 Direction count and computational cost

The complex schedule is most compact for repeated, balanced active exponents, including $(2, 2)$, $(2, 2, 2)$, $(3, 3)$, and $(4, 2)$. For $(2, 2, 2)$, for example, it uses 9 directions rather than the 13 produced by coalesced real polarization. This reduction controls the batch size of the directional jet, but not every component of execution time. Complex arithmetic, kernel launch overhead, memory traffic, and the backward pass can dominate for small networks or small batches. The rank theorem should therefore be read as an exact algebraic complexity result, not as a hardware-independent speedup.

7.2 Interpretation of the PINN study

Under the frozen `jsc_v2` protocol, Complex sinh has the lowest mean error in all 12 benchmark settings. The strongest separation occurs in the two-dimensional fourth- and sixth-order polyharmonic problems and in the complex-valued Maxwell problem. The three-dimensional sixth-order problem is a necessary counterweight: all methods remain near unit relative error, and the difference between Complex sinh and SIREN is practically negligible.

The comparison does not isolate one causal mechanism. A complex model changes the value representation, initialization, parameter count, and throughput at the same time. Literal width $H = 128$ gives the scalar Complex sinh network roughly twice as many real degrees of freedom as SIREN, whereas MscaleDNN-2-sin has more degrees of freedom but completes fewer steps. The loss weights and optimizer are shared rather than tuned separately for every architecture. The reported result is therefore a controlled comparison of complete methods under one compute protocol, not proof that complex parameters are intrinsically superior.

7.3 Operator scope and stochastic alternatives

Square-free patterns attain $R_{\mathbb{C}} = 2^{p-1}$, so complex Waring gives no direction-count reduction. Operator sums can be expanded and evaluated monomial by monomial, as in the polyharmonic experiment, but this procedure is not jointly rank-optimal for the full sum. Trace-collapsing approaches [6, 7] can be preferable for Laplacian-only workloads.

The Stochastic Taylor Derivative Estimator [5] solves a strictly broader problem – arbitrary contraction of the order-k derivative tensor – via stochastic sparse jets driven by integer partitions. For one fixed multi-index ν, our schedule is algebraically exact, whereas STDE is unbiased with variance controlled by its sampling budget. The approaches are complementary: stochastic estimation addresses large contractions, while Waring provides a minimal deterministic formula for one monomial derivative.

8 Conclusion

We have shown that the construction of a directional schedule for one mixed partial derivative $\partial^\nu u$ of order p is exactly the problem of a Waring decomposition of the monomial z^a. The classical Carlini–Catalisano–Geramita rank theorem and its roots-of-unity construction yield an explicit, rank-optimal directional schedule over $\mathbb{C}$. Combined with batched Taylor-mode propagation and a custom vector–Jacobian product, it gives an algebraically exact, parameter-differentiable backend for high-order neural surrogates.

The method is at its most useful for repeated-index multi-indices, where the complex Waring rank is materially smaller than the coalesced real polarization count. For square-free patterns it offers no rank advantage; for operator sums it is applied term by term without a global optimality claim. The same entire-on-$\mathbb{C}$ implementation supports complex-parameter sinh PINNs. In the validated five-seed application study, that complete method gives the lowest mean error on all 12 tested settings, but all methods saturate on the hardest three-dimensional sixth-order case. These results support the method as a useful deterministic component for repeated high-order derivatives and complex-valued PDE surrogates, while preserving the need for operator-specific methods in broader contractions.

Code and data availability

The implementation, executable verification tests, experiment configurations, validated per-seed result bundles, and table/figure generators are available in the public apolarity repository. Each formal result records the protocol identifier, source commit, hardware and software environment, random seed, architecture metadata, completed steps, and evaluation rule.

A Proofs of directional schedule identities

A.1 Roots-of-unity filter

Lemma A.1. *For any positive integer m and integer k, $\sum_{\zeta:\,\zeta^m=1} \zeta^k = m\,\mathbf{1}_{\{m|k\}}$.*

Proof. If $m \mid k$ every term is 1; otherwise the sum is a non-trivial geometric series in the primitive m-th root of unity, which sums to zero. □

A.2 Proof of Theorem 3.6

We verify condition (2) of Theorem 3.1: namely that the schedule (9) satisfies $p!\,z^a = \sum_\zeta c_\zeta (v_\zeta \cdot z)^p$ in $\mathbb{C}[z]_p$.

Expanding $(v_\zeta \cdot z)^p$ by the multinomial theorem and collecting the coefficient of z^β (with $|\beta| = p$):

$$[z^\beta] \sum_\zeta c_\zeta (v_\zeta \cdot z)^p = \frac{\nu!}{\prod_{j=1}^n m_j} \binom{p}{\beta} \prod_{j=1}^{n} \Big(\sum_{\zeta_j^{m_j}=1} \zeta_j^{1+\beta_j} \Big).$$

By Lemma A.1, each inner sum equals m_j iff $m_j \mid 1 + \beta_j$, i.e. $\beta_j \equiv a_j \pmod{m_j}$, and equals zero otherwise.

A short counting argument shows that the unique feasible joint assignment is $\beta = a$. Indeed, the congruence forces $\beta_j = a_j + t_j(a_j+1)$ for $j \geq 1$ with $t_j \geq 0$. If some $t_j > 0$, then

$$\beta_0 = a_0 - \sum_{j=1}^{n} t_j(a_j+1) < 0,$$

because $a_j \geq a_0$, impossible. Hence all $t_j = 0$ and $\beta = a$. With $\beta = a$ all the m_j factors cancel and the leading constant becomes $\nu!\binom{p}{a} = p!$, giving $[z^a]\sum_\zeta c_\zeta (v_\zeta \cdot z)^p = p!$ and all other coefficients vanish. □

A.3 Apolar pairing alternative

The Waring rank $R_{\mathbb{C}}(F)$ of a homogeneous polynomial $F \in \mathbb{C}[z]_p$ admits the apolar characterization $R_{\mathbb{C}}(F) = \min\{\deg X : X \subset \mathbb{P}^n \text{ a finite reduced scheme apolar to } F\}$, where X is apolar to F iff $I_X \subset F^\perp$ and $F^\perp$ is the apolar (Macaulay inverse) ideal of F. For $F = z^a$, $F^\perp = (\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1})$, which is a complete intersection. The Hilbert function of the quotient $\mathbb{C}[\partial]/F^\perp$ gives the lower bound matching (8); details in [13, 14].

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